MCQs: MEDICAL GENETICS CHAPTER 24: UNIVERSITY PRESSS

ANSWER ALL QUESTIONS

1. What is the term used to refer to the full set of chromosomes from an individual?

A. Genome

B. **Karyotype**

C. Chromosome set

D. Gene map

2. Which syndromes are associated with an additional copy of chromosome 21?

A. Edwards syndrome

B. Patau syndrome

C. **Down syndrome**

D. Warkany syndrome 2

3. Which condition results from a trisomy of chromosome 18?

A. Down syndrome

B. Patau syndrome

C. **Edwards syndrome**

D. Turner syndrome

4. Which type of chromosomal abnormality involves gaining two extra chromosomes?

A. Trisomy

B. **Tetrasomy**

C. Pentasomy

D. Nullisomy

5. Which of the following describes Pentasomy?

A. Loss of one chromosome

B. Gain of one extra chromosome

C. Gain of two extra chromosomes

D. **Gain of three extra chromosomes**

6. What is the result of a complete loss of a pair of homologous chromosomes?

A. Trisomy

B. Tetrasomy

C. Pentasomy

D. **Nullisomy**

7. Which term refers to an organism with a single set of chromosomes?

A. Diploid

B. Polyploid

C. **Monoploid**

D. Euploid

8. Polyploidy refers to an organism with:

A. A single set of chromosomes

B. Two sets of chromosomes

C. **Three or more sets of chromosomes**

D. A missing set of chromosomes

9. Which chromosomal rearrangement involves the duplication of a chromosomal segment?

A. Deletion

B. **Duplication**

C. Inversion

D. Translocation

10. Which condition is caused by duplication of the gene encoding peripheral myelin protein 22 (PMP22) on chromosome 17?

A. Down syndrome

B. **Charcot-Marie-Tooth disease type 1A**

C. Edwards syndrome

D. Patau syndrome

11. What type of chromosomal rearrangement involves the exchange of segments between two different chromosomes?

A. Inversion

B. Duplication

C. **Reciprocal translocation**

D. Ring Formation

12. Which chromosomal abnormality only occurs with chromosomes 13, 14, 15, 21, and 22?

A. Inversion

B. Isochromosome

C. Ring formation

D. **Robertsonian translocation**

13. Which chromosomal abnormality occurs when a chromosome segment is broken, inverted, and reattached?

A. Duplication

B. **Inversion**

C. Translocation

D. Deletion

14. Which chromosomal rearrangement involves a portion of one chromosome being deleted from its normal place and inserted into another chromosome?

A. Inversion

B. Duplication

C. **Translocation**

D. Insertion

15. An isochromosome forms when:

A. **One arm of the chromosome is missing, and the remaining arm is duplicated**

B. Both arms of a chromosome are deleted

C. A chromosome is split into two identical chromosomes

D. A chromosome forms a ring structure

16. What type of chromosomal abnormality results in the formation of a hybrid gene product? A. Duplication

B. **Translocation**

C. Deletion

D. Inversion

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17. What is a diagram showing the ancestral relationships and transmission of genetic traits over several generations in a family called?

A. Gene map

B. **Pedigree**

C. Family tree

D. Karyotype

18. In a pedigree chart, what shape represents a female?

A. Square

B. **Circle**

C. Diamond

D. Triangle

19. What shape is used to represent a male in a pedigree chart?

A. Circle

B. **Square**

C. Diamond

D. Triangle

20. If the sex of an individual is unknown in a pedigree, what shape is used?

A. Circle

B. Square

C. **Diamond**

D. Triangle

21. In a pedigree, how are parents connected?

A. By diagonal lines

B. **By a horizontal line**

C. By a vertical line

D. By a dotted line

22. What connects the parents to their offspring in a pedigree chart?

A. Horizontal line

B. **Vertical line**

C. Diagonal line

D. Curved line

23. How are siblings arranged in a pedigree?

A. From right to left according to birth order

B. **From left to right according to birth order**

C. From top to bottom according to birth order

D. In a circular pattern

24. How are twins represented in a pedigree chart?

A. By a vertical line connecting the parents

B. By horizontal lines connecting the siblings

C. **By diagonal lines connecting the twins**

D. By a circle and a square connected by a straight line

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25. Which term is used specifically for a male whose phenotype is of interest in a pedigree?

A. Proposita

B. Proband

C. **Propositus**

D. Offspring

26. Which term is used specifically for a female whose phenotype is of interest in a pedigree?

A **Proposita**

B. Propositus

C. Proband

D. Ancestor

27. In a pedigree chart, how is the proband indicated?

A. By a circle

B. By a square

C. **By an arrow pointing to the individual**

D. By a diamond

29. During which phase of mitosis does the nuclear envelope break down?

A. **Prophase**

B. Metaphase

C. Anaphase

D. Telophase

40. Cytokinesis is referred to as_______-

A. Division of the nucleus

B. Division of the chromosomes

C. **Division of the cytoplasm**

D. Division of the centromere

41. Which phase of interphase follows the S phase?

A. G1

B. **G2**

C. G3

D. M

42. What is the role of centrioles in mitosis?

A. Replication of DNA

B. Formation of the nuclear envelope

C. **Initiation and organization of the mitotic apparatus**

D. Cytokinesis

42. One feature of the telophase is _________

A. Chromosomes align at the spindle equator

B. **Nuclear membranes reform around daughter nuclei**

C. Sister chromatids separate

D. Chromosomes condense and become visible

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43. Which cell cycle phase is characterized by cell growth and production of proteins and organelles?

A. **G1**

B. S

C. G2

D. D. M

44. During which phase do the chromosomes relax into their extended phase?

A. Prophase

B. Metaphase

C. Anaphase

D. **Telophase**

45. What term describes the two identical DNA molecules produced during the S phase?

A. Chromosomes

B. **Sister chromatids**

C. Centromeres

D. Nucleosomes

46. During the telophase, the _______disappears

A. Nuclear membrane

B. **Spindle fibers**

C. Nucleolus

D. Centriole.

47. What is the primary function of the G2 phase?

A. DNA replication

B. Chromosome alignment

C. **Preparation for mitosis**

D. Cytokinesis

48. What phase immediately precedes anaphase in mitosis?

A. Prophase

B. **Metaphase**

C. Telophase

D. Cytokinesis

49. What is the main event of metaphase in mitosis?

A. Chromosome condensation

B. **Chromosome alignment at the spindle equator**

C. Separation of sister chromatids

D. Reformation of nuclear membranes

50. During which phase do progeny centrioles move apart?

A. **Prophase**

B. Metaphase

C. Anaphase

D. Telophase